Name _____ Period ____

Numerical Operations

Your Tasks (Mark these off as you go)

- ☐ Create the NumericalOperations driver class
- ☐ Evaluate and print numeric expressions
- ☐ Predict integer division and modulus results
- ☐ Trace unary operators and apply compound assignments
- ☐ Predict results involving mixed data types and casting
- ☐ Reverse a four-digit number using / and %

☐ Create the NumericalOperations driver class

Consider a class file called NumericalOperations.java. In the space below, write code that could be used to declare the class and the main method.

☐ Evaluate and print numeric expressions

Write code to complete the following:

- Create three String variables and assign each equation label shown below to a different variable.

  "Problem 1: 79 + 3 * (4 + 82 - 68) - 7 + 19 = "
  "Problem 2: (179 + 21 + 10) / 7 + 181 = "
  "Problem 3: 10389 * 56 * 11 + 2246 = "

- Declare and initialize three int variables on a single line. Assign the result of each expression above to a different variable.
- Concatenate each result to its corresponding equation label.
- Print each equation label and its corresponding result to the console.

SC: I can use arithmetic operators, mixed data types, constants, and Math class methods to perform computations.

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□ Predict integer division and modulus results

Refer to the following code. Remember that int division discards the decimal portion and % returns the remainder.

```
int dividend = 12, divisor = 4, quotient = 0, remainder = 0;
int dividend2 = 13, divisor2 = 3, quotient2 = 0, remainder2 = 0;
quotient = dividend / divisor;
remainder = dividend % divisor;
quotient2 = dividend2 / divisor2;
remainder2 = dividend2 % divisor2;
```

Write the exact output for each statement.

`System.out.println(quotient);`	
`System.out.println(remainder);`	
`System.out.println(quotient2);`	
`System.out.println(remainder2);`	

□ Predict results involving mixed data types and casting

Work through each expression in order. Remember that int / int uses integer division; casting one operand to double changes the division to floating-point division.

```
double d1 = 6.8;
int i1 = 12;
int i2 = 18;
```

Expression	Predicted result
24.0 * (i1 / i2) + 1	
24.0 * ((double) i1 / i2) + 1	
i1 + d1 * 2	
i1 + (int) d1 * 2	

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☐ Trace prefix and postfix increment/decrement operators

Trace the value of x from one line to the next. Write the exact output produced by each print statement.

```
int x = 4;
System.out.println(x++);
System.out.println(++x);
System.out.println(x--);
System.out.println(--x);
```

`System.out.println(x++);`	
`System.out.println(++x);`	
`System.out.println(x--);`	
`System.out.println(--x);`	

☐ Apply compound assignment operators

For each row, write the equivalent compound assignment statement and the final value of x.

Starting code	**Equivalent statement and final value**
int x = 10; x = x + 6;	
int x = 24; x = x / 4;	
int x = 17; x = x % 5;	
int x = 8; x = x * 3;	

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□ Reverse a four-digit number using / and %

Write code that prints any positive four-digit int in reverse order. Use only / and %; do not convert the number to a String. Print each digit as it is extracted so zeros are preserved. Consider the int below:

int number = 1234;

When your code runs, "4321" should print to the console.

Below are more examples:

Input	Expected output
int n1 = 3455;	5543
int n2 = 8767;	7678
int n3 = 1200;	0021

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